ACG RACING DEPARTMENT

All Conditions Athletes

Palm Springs • February 2026

Over 25 professional trail runners from the United States, France, Japan, Korea, and beyond arrived in Palm Springs for the ACG Racing Department's winter performance camp. Rain and shine, they clocked some of their biggest volume of the year on the rugged desert terrain of the Coachella Valley — surrounded by scientists, designers, and engineers who came to listen, measure, and learn.

25+	5	10	4
ATHLETES	DAYS	COUNTRIES	STUDIES

From the **Nike Sport Research Lab × ACG** — API team leading the performance science agenda

THE SCIENCE

Four studies. One question: who is the trail athlete?

The API team set up a mobile performance lab at the Ritz Carlton Rancho Mirage and ran four concurrent research protocols throughout the week. The aim wasn't just to test — it was to build a **comprehensive physiological profile** of the trail runner.

Speed × Gradient

How does **lactate threshold shift** across inclines? Athletes ran treadmill intervals at −5%, 0%, +5%, and +15% grade while blood lactate, muscle oxygen, and VO_2 were measured.

Muscle Oxidative Capacity

NIRS sensors and blood flow occlusion cuffs measured how quickly the **quads and calves consume and recover oxygen** — a window into muscle resilience and fatigue resistance.

Hemoglobin Mass

By tagging hemoglobin with trace carbon monoxide, the team mapped **total blood volume, red cell volume, and oxygen-carrying capacity**.

Cognitive Assessment

A performance neuroscientist explored how each athlete **approaches their race mentally** — how cognitive decline shapes dark hours of pain and duration.

THE MIND

Cognitive demands at the edge of fatigue

Trail running isn't just physical. At hour 15, fatigued and deep in the mountains, athletes are still making **split-second decisions** on technical terrain. The camp addressed this head-on: morning mindfulness sessions with a performance neuroscientist and clinical psychologist explored breathwork, sensory grounding, and visualization.

> *"The stoke was high from day one and continued throughout. The athletes signed up for nearly every learning opportunity they could. Safe to say, everyone wanted more."*
>
> CAMP DISPATCH

Rather than wiring athletes with EEG headsets, the neuroscientist sat with each athlete individually — exploring how they experience cognitive decline during racing, what strategies they use to stay sharp at 3am, and how deeply scrambled their brain feels after a 100K.

THE PRODUCT

This athlete dresses in a system

One of the week's clearest signals: trail runners think about their gear as a **complete system**. From shoes to vests to shirts to hats — everything must work together across hours of variable terrain, weather, and effort.

KIGERFLY | PICKLEJUICE | ZEGAMA | ULTRAFLY | TRAIL VESTS | WIND SHELLS | TRAIL SHORTS | RUNNING CAPS | COMPRESSION TIGHTS

THE SIGNALS

What we heard, saw, and measured

01

Gradient is the variable that changes everything

Speed, economy, and lactate response all shift dramatically with incline. Flat-ground physiology doesn't predict mountain performance. Athletes need gradient-specific pacing maps.

02

The calves are the canary

Calf muscle oxidative capacity and fatigue resistance may be the most underrated predictor of trail performance. Athletes who retain calf function deep into races hold their speed longer.

03

Cognition is the invisible limiter

The brain fatigues alongside the body. Decision-making degrades over hours, and the stakes on technical terrain are real. Extending cognitive sharpness is a competitive frontier.

04

These athletes are one step from survival

Ultra-long durations, extreme elevation, unpredictable weather, unstable footing. Trail running pushes closer to human survival limits than most sports.

THE ROSTER

ACG Racing Department 2026

From vertical kilometer specialists to 100-mile ultramarathon runners. The global team that came to the desert.

Adam Merry · Addie Bracy · Anthony Costales · Bailey Kowalczyk · Baptiste Coatantiec · Caleb Olson · Cesare Maestri · Christian Allen · Daniella Oemus · Drew Holmen · Erin Clark · Gabriella Lasalle · Hiroki Kai · Hyunji Kang · Hyunjin Noh · Jennifer Lichter · Lauren Gregory · Liam Meirow · Marie Nivet · Minchul Ko · Rachel Drake · Rebecca Kohne · Riley Brady · Ross Gollan · Shea Aquilano · Takako Takamura · Tyler Green

THE SCIENCE TEAM

Brett Kirby, PhD — NSRL · Brad Winn, MS — NSRL · Kat Boere, PhD Cand. — UVic/NSRL · Emily Farina, PhD — NSRL · Laura Hutchinson, PhD — NSRL · Brad Wilkins, PhD — Oregon · Mira Schoeberlein, PhD Cand. — Oregon · Emma Richardson, MS — Oregon · Carsten Lundby, PhD — Copenhagen · Vanessa Zoras, MS, RD — CSI · Tina Bryson, PhD, LCSW — Mental Health

FIRESIDE SEMINARS

MONDAY
Adaptations from Physiological Stress
Carsten Lundby, PhD

TUESDAY
Heat Storage & Cooling Strategies
Brad Wilkins, PhD

WEDNESDAY
Fluid & Fuel for the Trail
Vanessa Zoras, RD

THURSDAY
Eccentric Muscle Work: Optimize or Adapt?
Brett Kirby, PhD

Where science met the mountain — and where the mountain started talking back.

Five days in the desert. Four research protocols. 25+ athletes from around the world. The ACG Team Camp was a first-of-its-kind convergence of performance science, product development, and athlete voice — and it set the template for how the API team will embed with athletes in the field going forward.